Rational Number Project

Initial Fraction Ideas Lesson 6: Overview	Materials
Students observe with circles that as the unit is divided into more and more equal parts, the unit parts become smaller.	• Fraction Circles for students and teacher • Student Pages A and B

Teaching Actions

Warm Up

Show this fraction with your fraction circles using two different units or wholes. Then draw pictures for your display: $\frac{3}{4}$

Large Group Introduction

1. Start the lesson by reviewing ordering of whole numbers. For example, ask a student to select the greater of these 2 numbers, 720 or 702, and to explain his/her strategy for doing so.

2. Give another example using a context. José earns $42,175 a year. Mara earns $51,275 a year. Who earns more?

3. Introduce the idea of ordering fractions with this example. Draw two whole circles. On one circle draw in fourths. In the other draw in eighths.

4. Which circle has the larger parts? Why is this so? So which is bigger one of the four equal parts or one of the eight equal parts?

5. Explain: Let's develop this idea between number of parts and size of parts using Student Page A.

6. Ask students to use their fraction circles as you work together; name the black circle as the unit.

7. Ask: How many brown pieces cover the whole circle? How many orange? Which color takes more pieces to cover the whole unit? Which

Comments

To think quantitatively about fractions, students should know something about the relative size of fractions. Lesson 6 is the first of several lessons to help students construct informal strategies for ordering fractions. At Level 1, we want to provide the concrete experiences that students need if they are ever to reason intuitively about fraction symbols.

Activities in this lesson will lead students to reason, for example, that 1/4>1/8 because if you divide a circle into 8 equal parts, the parts will be smaller than if you divide the same unit into 4 equal parts.

This notion of more and greater can lead to misunderstandings. Some students may want to say that 1/8 > 1/4 because 8 > 4 or because with eighths, you have more pieces than you do with fourths. Whole-number reasoning has a strong influence on how children think about fractions.

Children need to be reminded that to compare fractions, we look at the "size of piece," not the "number of pieces".

Teaching Actions

color has the smaller pieces?

8. Record that information in a chart.

Color	How many cover 1 circle	Which color takes more…	Which color has smaller…
1. Brown	3		
Orange	5	√	√
2.			

12. When completed, ask students if they see any patterns between the number of pieces to fill the whole unit and the size of the pieces.

13. As a group, write a rule similar to either of these:
    - As the number of pieces needed to fill the whole decreases, the size of each piece gets larger.
    - As the number of pieces needed to fill the whole increases, the size of each piece gets smaller.

14. Once the rule is generated use it in examples without the circular pieces.

Examples:
20 purples = 1 whole
80 greens = 1 whole
Which is larger, 1 purple or 1 green?

Small Group/Partner Work

15. Assign Practice Pages B to reinforce the day's lesson.

Wrap Up

18. Ask students to picture the whole circle in their mind. What color pieces would divide the circle into twelfths? (red). Now picture the color pieces that would cover the whole circle into 3 equal parts (brown). Which piece is bigger red or brown? Why does that make sense?

19. Repeat for blue vs yellow; orange vs red. Then ask:

Comments

Teaching Actions	Comments
which is bigger and why: ½ or ¼? 16. Conclude by asking: Does more always mean less with fractions? Give this example: Imagine that it takes 10 maroon pieces to cover the whole circle. Which is smaller, 2 maroon pieces or 3 maroon pieces? How do you know? 17. Ask: How is this example different from all the rest we've talked about today? 20. Ask which is bigger: ¾ or ¼? What do you picture in your mind to answer this question?	This lesson helps students to compare unit fractions. These are fractions with one in the numerator. The term "unit" is used differently here than when you call the whole circle the unit. This can be confusing for students so there is not need to label the fractions as unit fractions. In the next lesson students compare fractions with the same numerators other than one.

Translations

- Real life to picture to verbal
- Manipulative to written symbols to verbal
- Written symbols to manipulative
- Real life to manipulative to pictures

Show this fraction with your fraction circles using two different units. Then draw pictures for your display:

$$\frac{3}{4}$$

Directions: **Use fraction circles to fill in the table.**

Color	How many cover 1 whole circle?	Which color takes MORE pieces to cover 1 whole?	Which color has SMALLER pieces?
1. Brown	**3**		
Orange	**5**	√	√
2. Orange			
White			
3. Purple			
White			
4. Gray			
Green			
5. White			
Green			
6. Orange			
Purple			
7. Gray			
Brown			
8. Brown			
Green			

Comparing Fractions

1. Use the whole circle as your unit. Show $\frac{1}{3}$ and $\frac{1}{2}$. Which fraction is the smaller of the two?

2. Use the whole circle as your unit. Show $\frac{1}{5}$ and $\frac{1}{10}$. Which fraction is the larger of the two?

3. Use the whole circle as your unit. Show $\frac{1}{6}$ and $\frac{1}{3}$. Which fraction is the smaller of the two?

4. Use the whole circle as your unit. Show $\frac{1}{12}$ and $\frac{1}{9}$. Which fraction is the larger of the two?

5. Imagine the whole circle is a pizza. You cut the pizza into 4 equal parts. How much of the pizza is one slice? Would you get more or less pizza if you divided it into 8 equal parts? Draw pictures to show how you answered the questions.